# Heurémen

QUANTUM IDLE AUDIT

## QIA-2026-001

Opened 22 October 2026 · issued 23 October 2026 · ibm_fez, ibm_kingston, ibm_marrakesh, ibm_phoenix, ibm_pittsburgh, ibm_boston

### Deviations from stated figures

| # | IBM stated figure | Chip | IBM's own documentation | Delivered [95 % interval where given] |
|---|---|---|---|---|
| 1 | Coupling `zz`, edge by edge: shift of an idle qubit when one neighbour changes state<br>IBM states: 11.6 to 58.8 kHz, median 28.3 (37 pairs); three more stated 0.00 | phoenix; 22 and 23 Oct | *"Unfortunately, unwanted interactions between qubits can lead to coherent errors on idling qubits." (IBM, error mitigation guide); no zz figure is documented* | Two runs on the same pairs, fixed before each job. Beat beside a superposed neighbour: median 12.9 kHz, slope on stated 0.19 [0.01, 0.38]. Shift, neighbour 0 then 1: 0.49 [0.45, 0.53] of stated, slope −0.06 [−0.63, 0.32] (0.19 [0.04, 0.37] without qubit 41's two slipped fits)<br>**Stated zz does not predict the coupling (slope 0.38 or less in both runs; a convention factor leaves it near 1)**<br>One chip. Same code on fez: slopes 1.00 [0.89, 1.14] and 0.92 [0.83, 1.02]. A weak dependence remains (beat rank correlation 0.39 [0.05, 0.66]). Cause not identified |
| 2 | `prob_meas1_prep0`, P(1 \| 0)<br>IBM states: 0 on 14 qubits; every marrakesh value in that record is a multiple of 1/1,024, and the 95 % bound of 0 of 1,024 is 0.29 % | marrakesh, one calibration session 22 Oct 16:25 EDT (carried in records stamped 19:23 and 20:14) | *"The latter is a measurement of both initialization error and the procedure of discriminating between the \|0⟩ and \|1⟩ states." (IBM, QPU information guide)* | Read alone, 20,000 shots each: 9 of 14 above 0.29 %, median 0.56 %; read with every qubit, two jobs: 11 of 14 above 0.29 % in each, medians 0.73 % and 0.66 %<br>**Stated zero on 14 qubits in one session; delivered above what a zero allows on 9 to 11 of them.**<br>Same evening, 3.9 h after IBM's measurement; isolated and all-qubit reads agree (difference −0.0024 [−0.0048, +0.0001]); every qubit on the chip read, 20,000 shots per state. Fez: 1 of 2 stated zeros above; kingston: 0 of 1 |

### Record notes

| # | Record | Chip | IBM's own documentation | What IBM's record shows |
|---|---|---|---|---|
| 3 | T1 dates | marrakesh, record stamped 21 Oct 22:11 EDT | *"The QPU-reported properties are updated once a day, but the system may drift faster than the time between updates." (IBM, real-time benchmarking tutorial)* | one T1 value dated 14 Sep (895 h earlier); 14 of 154 dated more than 24 h earlier |
| 4 | T1 and T2 dates | kingston, records stamped 26 to 28 Sep | *"The QPU-reported properties are updated once a day, but the system may drift faster than the time between updates." (IBM, real-time benchmarking tutorial)* | one T2 value dated 2,332 h (97 days) earlier; 4 to 6 of 155 T1 values over 24 h old |
| 5 | T1 and T2 dates | marrakesh, records stamped 26 to 28 Sep | *"The QPU-reported properties are updated once a day, but the system may drift faster than the time between updates." (IBM, real-time benchmarking tutorial)* | 9 to 10 of 154 T1 values over 24 h old, oldest 340 h (14 days); one T2 value 734 h (31 days) old |

| | | | | |
|---|---|---|---|---|
| 6 | T1 and T2 under a new stamp | boston, fez, kingston, marrakesh, phoenix, pittsburgh; 16 Sep to 23 Oct | *"last_update_date is updated whenever a calibration occurs, independently of the per-property benchmark timestamps." (IBM, calibration-jobs guide)* | the stamp IBM's platform labels "last calibration" moves while every T1 and T2 entry keeps its value and measurement date (4,204 of 4,204 entries, 14 record pairs up to 50 h apart; the 3 other consecutive pairs we hold, 71 to 765 h apart, show them remeasured), so those figures are older than the stamp: boston T1 and T2 measured by 21 Sep 11:02 EDT shown under stamps of 22, 23 and 25 Sep; phoenix T1 and T2 unchanged under three stamps, 5 Oct 09:44, 6 Oct 10:45 and 12:25; fez under four stamps, 22 Oct 18:17 to 23 Oct 04:38; kingston T1 and T2 measured by 22 Oct 05:02 EDT shown under the 23 Oct 04:55 stamp |
| 7 | Coupling `zz` on five live edges | fez 13–14, 39–53; phoenix 100–101, 70–80, 86–96; records in force 21 and 22 Oct | *"Unfortunately, unwanted interactions between qubits can lead to coherent errors on idling qubits." (IBM, error mitigation guide)* | zz = 0.00 on edges whose CZ gates the same record calibrates (error fez 0.0019, 0.0031; phoenix 0.029, 0.0011, 0.0011); IBM's earlier records give 7.45 and 2.89 kHz (fez, 26 to 29 Sep) and 17.46, 14.13 and 13.68 kHz (phoenix, 5 and 6 Oct); delivered 8.28 and 2.85 kHz (fez, direct shift, copies agree) and 14.21, 15.36 and 10.23 kHz (phoenix, direct shift, copies agree). A model built from the record in force sets these couplings to zero |
| 8 | `init_error` | fez / kingston / marrakesh; marrakesh record of 21 Oct 21:08 EDT | *"perform daily benchmarking" (IBM, calibration-jobs guide)* | absent for 40 / 31 / 9 qubits; 18 / 17 / 19 of the present values over 7 days old; 26 to 28 Sep records (kingston / marrakesh): absent for 31 / 9, oldest 57 and 61 days |
| 9 | Readout error between cycles | fez / kingston / marrakesh | *"The QPU-reported properties are updated once a day, but the system may drift faster than the time between updates." (IBM, real-time benchmarking tutorial)* | per-qubit values change by a median 20 % (fez), 16 to 30 % (kingston), 29 to 62 % (marrakesh) between consecutive cycles 2 to 15 h apart (October records), and 24 to 27 % (fez), 26 to 27 % (kingston), 35 to 36 % (marrakesh) between daily September records 12 to 28 h apart; IBM's counts are multiples of 1/4096 (4096 shots per state), where sampling alone would give about 9 / 10 / 8 %; marrakesh's records of 22 and 23 Oct are multiples of 1/1024 (1024 shots), where sampling alone gives about twice that |
| 10 | Stale values | boston, fez, kingston, marrakesh, phoenix, pittsburgh; 66 records, 16 Sep to 22 Oct | *"If the benchmarking of a qubit or edge does not succeed over the course of several days … the reported error value is considered stale and will be reported as 1." (IBM, QPU information guide)* | the stale flag appears on 2,101 gate_error entries and on no readout_error or init_error entry. init_error, an error value, is never flagged: 145 values shown more than 7 days after they were measured, oldest 84 days, on qubits whose readout was under 2 days old. T1 and T2 carry no flag: 6 T1 (oldest 42 days) and 7 T2 shown more than 7 days old, oldest T2 158 days (pittsburgh qubit 117, measured 12 Apr, shown under the 17 Sep stamp) |
| 11 | `readout_error` and its published parts | boston, fez, kingston, marrakesh, phoenix (none on pittsburgh); 66 records on six processors | *"The reported value is the average of these two errors." (IBM, QPU information guide)* | where readout_error and both parts carry one date (3,766 measurements) it is exactly their average; on 153 measurements (43 qubits) it is not, and in every one the published `prob_meas0_prep1` is older than the readout_error it should average into, by up to 97.5 h (18 off by more than 0.01, largest 0.157). 141 of the 153 are on qubits 16, 33, 50, … (every 17th). The transpiler uses readout_error (156 of 156 qubits on fez, kingston and marrakesh, 22 Oct) |

## Not in any stated figure

| # | Effect | IBM states | Delivered [95 % interval where given] |
|---|---|---|---|
| 12 | Phase kick on untouched qubits per mid-circuit readout | no field in the calibration record | fez −1.62 [−2.08, −1.18], kingston −2.86 [−3.55, −2.09], marrakesh −2.39 [−3.00, −1.72] degrees per readout (sizes 1.6 to 2.9); with 10 of the 39 read out, fez −0.17 [−0.32, −0.07] (22 Oct repeat, both copies)<br>39 of 156 read out, every untouched qubit within two couplings of a measured one; against an idle of the stated readout length, same job, two copies; default measure instruction (measure_2: marrakesh 0.51 to 0.58 of it; kingston 0.95 [0.75, 1.17] and 0.98 [0.77, 1.28] at 4 and 8 readouts, repeat, length ratio 0.63 excluded; fez cannot tell); repeated 22 Oct in the echo run's no-echo arms (−1.50, −3.24, −2.50); qubits whose idle phase turns the other way are kicked the same way |
| 13 | Coherence lost by untouched qubits next to a measured one, 4 and 8 readouts | no field in the calibration record | kingston +0.0056 and +0.0062; marrakesh +0.0087 and +0.0205 (all above zero with copies pooled; kingston at 4 readouts: neither copy alone resolves it); fez cannot tell at 2 readouts under the committed rule and is consistent with zero at 4 and 8<br>Only shots where every measured neighbour was verified in its \|0⟩ cluster on every readout (97 to 99 % of shots); a neighbour left in a level that reads as 0 is excluded by an X-flip check on marrakesh, to the extent the flip catches it; neighbours excited and misread as 0 (0.4 to 1.1 % of shots stuck in \|1⟩) are not excluded as the whole cost |
| 14 | Entanglement lost by untouched Bell pairs, 4 readouts | no field in the calibration record | Clean-shot loss: fez 0.0070 [0.0016, 0.0152], repeat 0.0051 [0.0005, 0.0126]; marrakesh 0.0093 [0.0005, 0.0155], repeat 0.0094 [0.0037, 0.0243]; kingston 0.0100 and, repeat, 0.0068 on clean shots (cannot tell under the committed rule), above zero on all shots in both. 39-readout minus 1-readout difference (the unbiased reading), 21 Oct: marrakesh 0.0088, kingston 0.0104, fez 0.0065 (cannot tell under the committed rule); repeat: above zero on kingston and marrakesh, consistent with zero on fez<br>Readout against equal idle, same job, two independent runs never pooled; keeping only shots with every measured neighbour verified in \|0⟩ can bias the loss upward, which the 39-minus-1 difference removes; in the control run's selection marrakesh is consistent with zero at 2 and 4 readouts (+0.0060, +0.0056), while fez reads +0.004 at 4 and kingston +0.010 and +0.009 at 2 and 4, above zero |
| 15 | Probe offset in weak-measurement circuits | no field in the calibration record | marrakesh q8, 22 Oct: +0.049 to +0.081 (±0.010) in three circuit variants; the standard noise model built from the calibration at firing gives +0.003 to +0.005, and q8's stated readout asymmetry accounts for +0.006; fez 11 Oct +0.062; 24 Sep +0.060, the mean of four jobs on marrakesh, kingston and fez<br>Post-selected probe in the box whose ideal value is 0; the model is built from IBM's calibration, not a figure IBM states. Probe split, 22 Oct: probe readout and state preparation consistent with zero on fez and marrakesh; the part carried by the controlled gate lies within what the stated CZ error allows if that error is coherent. Readout-corrected offsets stay above zero. Cause not identified |
| 16 | Idle decay is not single-exponential | one T1 per qubit, a single-exponential fit | 62 to 85 % of qubit-copies prepared in \|1⟩ read below the line from their own 300 µs level at 30 µs (26 jobs, 4 to 7 October), where a pure exponential gives 49 to 54 %<br>All qubits prepared in \|1⟩, read at 30 to 600 µs in one job; the job's own \|0⟩ and \|1⟩ circuits remove readout error. The test cannot separate a non-exponential decay from a per-qubit systematic common to the three copies; the data do not identify the cause |

Method: same quantity, same definition, against IBM's calibration record saved when the job ran. Medians over qubits, 95 % intervals; a deviation is listed only when IBM's figure falls outside our interval. Data and scripts: heuremen.org/experiments · doi:10.5281/zenodo.23169755.

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